Disruptors

Purpose

Like the fires breaking out in Little Cities or the UFO attacks in Sim City, at certain milestones, the game needs to “test” the player’s currently loaded habitat to break the linear progression.

Disruptors should test the player’s ability to strategically place plants and save points to make sure that they can sustain their habitat beyond the minimum requirements. It should prevent them from simply planting an excess of plants and making way for the herd early by exploiting dinopoints.

They are intended to add an element of unpredictability and challenge the players progression as the player can anticipate and respond and attempt to these events to ensure harmony in their habitats.

The disruptor weather handler also handles the rotation of regular rain events and other weather.

Mechanics

Changing Weather

Weather occurs on a timer with a probability.

Every X seconds the game does a check to see if a regular weather change occurs.

Currently there is ‘Sunny’ and ‘Rainy’ weather but the system should be left expandable

Weather Event	Example Probability
Sunny	80
Rainy	20

Event trigger

Disruptor events occur semi-randomly throughout gameplay with some condition checks. Disruptors do not trigger when there is only one dinosaur in the environment (the buddy).

Disruptors cannot ever kill or remove your buddy.
There is a timer that counts down until an event occurs and a random range multiplier.

The game will not repeatedly play the same event every event interval.

There is an event timer.
The event timer is modified by a random range threshold.

When the event timer counts down to zero, the game checks whether the conditions for each weather event are met and checks the probability of each occurring.

When it decides which occurs, a disruptor is initiated.

The probability is incremented as the player levels up and the timer is decremented.

Variable	Explanation	Value
EventTimer	When this timer hits zero, an event is rolled and potentially triggered.	1800
EventProbability	The probability that the event fires when the timer hits zero.	0.1
RandomRangeUpper	Upper random range multiplier for the timer.	1.5
RandomRangeLower	Lower random range multiplier for the timer.	0.5
ProbabilityLevelIncrement	How much the likelihood of a disruptor goes up when the player levels up.	0.10

TimerLevelDecrement	How much the timer goes down when the player levels up.	100

Climate events

General

The weather will force players to replace certain types of plants when certain conditions are met.

When the herd counts 2+ Adult Dinosaurs or above, every 30 minutes (multiplied by a random range 0.5-1.5) of accumulated in-game time.

Dinosaurs stop what they are doing and attempt to seek cover when event plays out.
Plants used as cover are not destroyed by the event.

These events last for about 60 seconds and have a timer.

Note:
Alternatively, we could set them up so that they trigger at about the 20+ minute or so mark and let them run for several minutes time to make them feel more real and encourage people to sign out for the day or to timelapse and finish it sooner.

Variable	Explanation	Value
NumberOfDinos	Number of dinosaurs required to activate the disruptor.	2
EarliestLifeStageActivated	The stage at which the disruptors activate for the level	Adult (should be able to pick from any life stage)

EventTime	The length of time the visual representation occurts	60

Rain and Storms

When the game determines that the storm disruptor will occur, a pre-storm interval will be triggered.
The regular rain event starts and is randomized between a minimum and maximum amount.
After the timer expires, the storm event proper begins.

The game should never abruptly transition from sun to a storm instantaneously.
Neither should it immediately transition from a storm to sunny.

The storm breaks/kills tagged (tall) player-planted vegetation, e.g. trees and knocks them over, leaving debris the player needs to clean up.

The sun becomes dim and the colours become slightly desaturated when the storm starts. The sky becomes thick with clouds, rain falls and a thick mist covers the level that drops visibility to a short distance (thick fog and a low culling distance to hide objects from view and obscure the environment). The player can hear the storm raging, trees groaning and eventually some bending and breaking before the storm lets up.

The storm will break a configurable percentage of trees and tall plants that are not being used as cover by dinosaurs during the event. This allows designers to be flexible about the damage level without a complicated system and potentially add a layer of modifiers after playtesting.

Variable	Explanation	Value
PercentageOfTreesToBreak	Percentage of non-cover trees to knock down during the event.	0.25
RainTimer	Base time for how long the regular rain lasts before the storm.	120

RainTimerMinModifier	Lower range of the rain timer modifier	0.75
RainTimerMaxModifier	Lower range of the rain timer modifier	1.25

Volcano event

The volcano culls/kills old/larger dinos (not buddy) and withers some plants the player needs to clean up.

The sun becomes dark as if eclipsed and a red-light filter covers the level.
The air becomes humid and very blurry as well as wavy as if the heat is causing steam to rise.
The player can hear an aggressive ambient drone to underline the beginning of the eruption.
The volcano in the background lights up, casting an ominous crimson glow on the level as lava spews from the top.

Dinosaurs run for cover and cower. The eldest dinosaurs that will die hide out of sight of the player and are deleted during the event.

The volcano will burn a configurable percentage of bushes and trees that are not being used as cover by dinosaurs during the event. This allows designers to be flexible about the damage level without a complicated system and potentially add a layer of modifiers after playtesting. The heat wave will also kill fungus that is not directly transferring nutrients to plants.
The volcano will kill a random number of dinos from a configurable range, but never the buddy. They are chosen by age - eldest first.

Variable	Explanation	Value
PercentageOfPlantsToBurn	Percentage of non-cover bushes to burn during the event.	0.10
PercentageOfFungiToBurn	Percentage of fungi that are not serving compost to burn during the event.	0.10
NumberOfDinosToKillUpper	Max Dinos the event will take	2

NumberOfDinosToKillLower	Minimum number of dinos the event will take.	0

Stretch Goals

Heatwave

The heat wave withers and kills smaller player-planted vegetation, e.g. ferns and mushrooms., leaving dead debris the player needs to clean up.

The sun becomes obscured as if eclipsed and the colours become oversaturated on the red channel.
The air becomes humid and slightly blurry, and a little wavy as if the heat is causing steam to rise. The player can hear cicadas and a slightly aggressive ambient drone to underline the hear.

The wave will burn a configurable percentage of bushes that are not being used as cover by dinosaurs during the event. This allows designers to be flexible about the damage level without a complicated system and potentially add a layer of modifiers after playtesting. The heat wave will also kill fungus that is not direcly transferring nutrients to plants

Variable	Explanation	Value
PercentageOfBushesToBurn	Percentage of non-cover bushes to burn during the event.	0.25
PercentageOfFungiToBurn	Percentage of fungi that are not serving compost to burn during the event.	0.25

T.Rex event

When the herd counts 2+ Adult Dinosaurs or above, every 30 minutes (multiplied by a random range 0.5-1.5) of accumulated in-game time, an adult T.Rex will spawn in the habitat out of sight of the player at a spawn point specified by designers. The player is alerted by the sound of stomping feet and loud roars (spatial audio).

The music changes from a default chill vibe to silence or else a tense monotone droning noise that instills terror in the player.

The Tutorial/Narrative System is temporarily disabled while the event is ongoing.

Triceratops call to each and raise alerts when the see the T-Rex (detection radius) and run to hide. The oldest Triceratops (non-buddy) stands still while the other dinosaurs (including your buddy) runs in the opposite direction of the T.Rex (locations can be predefined by design) until they get to an egress point (collider) along the edge of the map.
Then they play “sad” animation for the remainder of the event.

Then the T.Rex vs Prey sequence plays out as the T-Rex Approaches the oldest dinosaur and kills it.

Once the sequence is over, the game returns to its normal state.

The player is notified by In-Headset Notifications that a raid is occurring or has occurred while they are out of the headset.
The losses are explained briefly when the player drops in world by puck - ‘Two dinosaurs were lost in a T rex Raid’.

Variable	Explanation	Value
NumberOfDinos	Number of dinosaurs required to activate the disruptor.	2
EarliestLifeStageActivated	The stage at which the disruptors activate for the level	Adult (should be able to pick from any life stage)

Old notes

NOTES

For example, during a thunderstorm, players may need to create shelter for their dinosaurs who now all we seek them out at the same time or the player need to prepare extra access to water for a draught.

Dinosaurs will behave differently depending on the weather. For example, they may be more active during rain or seek shelter during thunderstorms.

The weather will affect the condition of the environment. Rain will make the ground look wet, while heat waves will dry it out.

Types

The following weather types will be included in the game:

Shelter

General

The player needs to protect the dinosaurs from dangers such as extreme weather and predators by providing enough shelter in the habitat.

Rocks

Small shelter.

Dead logs

Medium Shelter.

Trees

Large shelter.

We could discuss whether the above is deprecated as trees have the same function.

Event

Prompt

Generated image

Player action

Dinosaur egg is about to hatch

”Come meet your Dino…”

Interact with dino

Dinosaur reaching the next life stage

”Look, I’m growing…”

Interact with dino

Extreme weather, thunderstorm, too many predators…

”Dino is in danger / is scared/cold...”

Create shelters, and insert placeables (plants) that increase climate temperature

Harvest possibilities

”Dino is hungry...”

Create food sources, harvest...

”Your Dino needs attention - sad” or

”Your Dino needs attention - playful”

Interact with dino

Extreme weather conditions

”Your Dino is too hot...” - ”Your Dino is thirsty...”

Create more bodies of water, and insert placeables (plants) that decrease climate temperature

Weather	Effect	Player strategy
Rain	Rain will cause the environment to become wet and slippery. Dinosaurs will be less likely to move around quickly or engage in combat during a rainstorm. However, some herbivorous dinosaurs may be more active during rain because it helps them to find food.	Prepare food sources.
Thunderstorms	During thunderstorms, dinosaurs will take cover and seek shelter.	Prepare shelter.
Heat Waves	The temperature will be high, and dinosaurs will be more prone to exhaustion and dehydration. Players will need	Prepare water sources.

	to make sure that their dinosaurs have access to enough water during heat waves.	